Compton Scattering: Index Tracking Edition

Every Lorentz Index on Every γ-Matrix, Every Contraction Shown

PHYS 7502

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Part I
How Gamma Matrix Indices Work

1 Indices 101: What Lives Where

The Index Rules

A gamma matrix γ^μ carries **one free Lorentz index** μ (runs over 0,1,2,3) and **two hidden Dirac indices** (the row and column of the 4×4 matrix):

$$(\gamma^\mu)_{ab} \qquad \mu = 0, 1, 2, 3 \quad ; \quad a, b = 1, 2, 3, 4$$

The Lorentz index μ:

- Tells you *which* gamma matrix (γ^0, γ^1, γ^2, or γ^3).
- Can be **upper** (γ^μ) or **lower** (γ_μ). Related by the metric: $\gamma_\mu = g_{\mu\nu}\gamma^\nu$.
- A **free** index means it is not summed over — it labels a family of equations.
- A **repeated** index (one upper, one lower) means **sum over it** (Einstein convention).

The Dirac indices a, b:

- The row and column of the 4×4 matrix.
- Almost always suppressed (we just write γ^μ and let matrix multiplication handle them).
- Only written out explicitly when doing Casimir's trick.

2 Slash Notation and Index Content

What $\not{p}$ Means for Indices

$$\not{p} \equiv p_\mu \gamma^\mu = p_0\gamma^0 + p_1\gamma^1 + p_2\gamma^2 + p_3\gamma^3 \tag{1}$$

The index μ is **summed over** (repeated, one on p_μ and one on γ^μ). So $\not{p}$ has **no free Lorentz index** — it is a single 4×4 matrix, not four matrices.
Compare:

- γ^μ = four different matrices, labeled by the free index μ.
- $\not{p} = p_\mu\gamma^\mu$ = one matrix (the sum over all four).

3 How Contractions Work

Contraction = Summing a Repeated Index

When an index appears **once upper and once lower**, you sum over it. This is called **contraction**:

$$\gamma^\mu\gamma_\mu \;=\; \sum_{\mu=0}^{3}\gamma^\mu\gamma_\mu \;=\; \gamma^0\gamma_0+\gamma^1\gamma_1+\gamma^2\gamma_2+\gamma^3\gamma_3 \;=\; 4\cdot\mathbf{1}$$

Before contraction: γ^μ has free index μ (upper), γ_ν has free index ν (lower). They are independent.
Contraction means setting $\nu=\mu$ and summing: $\gamma^\mu\gamma_\mu$. The result has **no free Lorentz indices** — it is a single 4×4 matrix.

4 The Contraction Theorems with Explicit Indices

What the Contraction Theorems Actually Do

Each theorem takes an expression with a **contracted pair** ($\gamma^\mu\cdots\gamma_\mu$, summed over μ) and simplifies it.

Theorem 7: $\gamma^\mu \not{a}\,\gamma_\mu = -2\not{a}$
Written with all indices visible:

$$\underbrace{\gamma^\mu}_{\mu\text{ upper}}\;\underbrace{a_\nu\gamma^\nu}_{\not{a},\,\nu\text{ summed}}\;\underbrace{\gamma_\mu}_{\mu\text{ lower}} = -2\,a_\nu\gamma^\nu = -2\not{a}$$

The index μ is contracted (summed over). The index ν inside $\not{a}$ is an internal dummy — it is already summed over and is unaffected. The result has no free Lorentz indices, just like $\not{a}$ itself.

Theorem 8′: $\gamma^\mu \not{a}\not{b}\,\gamma_\mu = 4(a\cdot b)$

$$\underbrace{\gamma^\mu}_{\mu}\;\underbrace{a_\nu\gamma^\nu}_{\not{a}}\;\underbrace{b_\rho\gamma^\rho}_{\not{b}}\;\underbrace{\gamma_\mu}_{\mu} = 4\,a_\nu b^\nu\cdot\mathbf{1} = 4(a\cdot b)\cdot\mathbf{1}$$

The contracted μ eats two γ's and two slashed momenta, producing a **pure number** times the identity matrix.

Theorem 9: $\gamma^\mu \not{a}\not{b}\not{c}\,\gamma_\mu = -2\not{c}\not{b}\not{a}$
The contracted μ eats three slashed quantities and **reverses their order**.

Part II

The Compton Calculation with Full Index Tracking

5 The Amplitudes with Every Index Shown

Convention: $1 = e^-_{\text{in}}$, $2 = \gamma_{\text{in}}$, $3 = \gamma_{\text{out}}$, $4 = e^-_{\text{out}}$. All at $m = 0$.

$\mathcal{M}_s$ with Every Index

$$\mathcal{M}_s = \frac{-e^2}{s}\, \varepsilon_3^{*\alpha}\, \varepsilon_2^{\beta}\, \bar{u}(4) \underbrace{\gamma_\alpha}_{\substack{\text{contracts}\\ \text{with } \varepsilon_3^*}} \underbrace{(p_{1\nu}\gamma^\nu + p_{2\rho}\gamma^\rho)}_{\not{p}_1 + \not{p}_2} \underbrace{\gamma_\beta}_{\substack{\text{contracts}\\ \text{with } \varepsilon_2}} u(1) \tag{2}$$

Index accounting:

- α: free on $\varepsilon_3^{*\alpha}$, contracted with γ_α at Vertex B. **Summed.**
- β: free on ε_2^{β}, contracted with γ_β at Vertex A. **Summed.**
- ν, ρ: internal dummies inside $\not{p}_1 = p_{1\nu}\gamma^\nu$ and $\not{p}_2 = p_{2\rho}\gamma^\rho$. **Already summed.**
- No free Lorentz indices remain. $\mathcal{M}_s$ is a Lorentz scalar (a number, once spinor indices are contracted).

6 Forming $|\mathcal{M}_s|^2$: New Indices for $\mathcal{M}_s^*$

Why We Need New Index Names in $\mathcal{M}_s^*$

In $\mathcal{M}_s$, the indices α and β are already summed (contracted with the polarization vectors). When we form $|\mathcal{M}_s|^2 = \mathcal{M}_s \times \mathcal{M}_s^*$, we are multiplying two separate sums. We **must use different dummy index names** in $\mathcal{M}_s^*$ to avoid confusion:

$\mathcal{M}_s$ uses α, β $\qquad$ $\mathcal{M}_s^*$ uses ρ, σ

$$\mathcal{M}_s^* = \frac{-e^2}{s}\, \varepsilon_3^{\rho}\, \varepsilon_2^{*\sigma}\, \bar{u}(1)\, \gamma_\sigma\, (\not{p}_1 + \not{p}_2)\, \gamma_\rho\, u(4) \tag{3}$$

The product is:

$$|\mathcal{M}_s|^2 = \frac{e^4}{s^2} \underbrace{\varepsilon_3^{*\alpha}\varepsilon_3^{\rho}}_{\text{photon 3 pair}} \underbrace{\varepsilon_2^{\beta}\varepsilon_2^{*\sigma}}_{\text{photon 2 pair}} \times \Big[\bar{u}(4)\, \gamma_\alpha\, (\not{p}_1 + \not{p}_2)\, \gamma_\beta\, u(1)\Big] \Big[\bar{u}(1)\, \gamma_\sigma\, (\not{p}_1 + \not{p}_2)\, \gamma_\rho\, u(4)\Big] \tag{4}$$

At this stage we have **four free Lorentz indices**: α, ρ, β, σ. Each appears exactly twice (once in the polarization pair, once on a γ-matrix), so they are all summed.

7 Casimir's Trick: Spinor Indices Become a Trace

After applying completeness for particles 1 and 4 (as shown in the spinor completeness document):

$$\sum_{\text{spins}} = \text{Tr}\Big[\gamma_\alpha\,(\not{p}_1+\not{p}_2)\,\gamma_\beta\,\not{p}_1\,\gamma_\sigma\,(\not{p}_1+\not{p}_2)\,\gamma_\rho\,\not{p}_4\Big] \tag{5}$$

Reading the Trace: Where Each Index Came From

γ_α	from $\mathcal{M}_s$ Vertex B	will contract with $\varepsilon_3^{*\alpha}$
$(\not{p}_1+\not{p}_2)$	from $\mathcal{M}_s$ propagator	no free index
γ_β	from $\mathcal{M}_s$ Vertex A	will contract with ε_2^{β}
$\not{p}_1$	from completeness of particle 1	no free index
γ_σ	from $\mathcal{M}_s^*$ Vertex A	will contract with $\varepsilon_2^{*\sigma}$
$(\not{p}_1+\not{p}_2)$	from $\mathcal{M}_s^*$ propagator	no free index
γ_ρ	from $\mathcal{M}_s^*$ Vertex B	will contract with ε_3^{ρ}
$\not{p}_4$	from completeness of particle 4	no free index

Four γ's have free indices $(\alpha,\beta,\sigma,\rho)$. Four more are hidden inside the slashed momenta ($\not{p}_1$, $\not{p}_2$, $\not{p}_1$, $\not{p}_4$), each with an internally summed index. Total: a trace of 8 gamma matrices.

8 Photon Sums: Contracting the Free Indices

8.1 Photon 3 sum: α meets ρ

$$\sum_{\lambda_3} \varepsilon_3^{*\alpha}\varepsilon_3^{\rho} \;\longrightarrow\; -g^{\alpha\rho} \tag{6}$$

What $-g^{\alpha\rho}$ Does to the Trace

The metric $g^{\alpha\rho}$ **contracts** α with ρ. In the trace, γ_α is the 1st matrix and γ_ρ is the 7th. The contraction turns them into a **matched pair**:

$$\gamma_\alpha \;\cdots\; \gamma_\rho \quad \xrightarrow{g^{\alpha\rho}\text{ contracts}} \quad \gamma^\alpha \;\cdots\; \gamma_\alpha$$

The index α now appears once upper and once lower $\Rightarrow$ it is **summed over**. This is exactly the structure that the contraction theorems are designed to simplify.
Mechanically: $g^{\alpha\rho}$ raises one index. γ_α stays as is (lower α), while γ_ρ becomes $g^{\alpha\rho}\gamma_\rho = \gamma^\alpha$... but actually since $g^{\alpha\rho}\gamma_\rho = \gamma^\alpha$, the simpler way to see it: the contraction just means "wherever you see ρ, replace it with α and sum." The net effect is that γ_ρ becomes γ_α, giving the pair $\gamma^\alpha\cdots\gamma_\alpha$.

8.2 Photon 2 sum: β meets σ

$$\sum_{\lambda_2} \varepsilon_2^{\beta}\varepsilon_2^{*\sigma} \;\longrightarrow\; -g^{\beta\sigma} \tag{7}$$

Same story: contracts β with σ, turning γ_β and γ_σ into the matched pair $\gamma^\beta\cdots\gamma_\beta$.

8.3 Result after both photon sums

Two factors of (-1) give $+1$. The trace becomes:

Trace After All Contractions (All Indices Now Summed)

$$\boxed{\frac{1}{4}\sum |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\,\mathrm{Tr}\Big[\gamma^\alpha\,(\not{p}_1+\not{p}_2)\,\gamma^\beta\,\not{p}_1\,\gamma_\beta\,(\not{p}_1+\not{p}_2)\,\gamma_\alpha\,\not{p}_4\Big]} \tag{8}$$

Index status:

- α: upper on 1st γ, lower on 7th γ. **Contracted pair.**
- β: upper on 3rd γ, lower on 5th γ. **Contracted pair.**
- All other indices (inside the $\not{p}$'s) are internal dummies already summed.

There are **zero free indices**. The trace is a pure number.

9 Evaluating the Trace: Using the Contraction Theorems

Strategy: Work from the Inside Out

We have two contracted pairs: β (inner) and α (outer). The strategy is:

1. Contract the **inner** pair (β) first — this simplifies the middle of the trace.
2. Then contract the **outer** pair (α).

You must work inside-out because the contraction theorems require the contracted γ's to be the **outermost** pair of the substring they act on.

9.1 Step 1: Inner contraction (β)

Locate the β pair in the trace. Between them sits only $\not{p}_1$:

$$\cdots\ \gamma^\beta\ \underbrace{\not{p}_1}_{\text{1 slash}}\ \gamma_\beta\ \cdots$$

This matches **Theorem 7**: $\gamma^\beta \not{a} \gamma_\beta = -2\not{a}$.

Applying Theorem 7

$$\gamma^\beta\,\not{p}_1\,\gamma_\beta = -2\not{p}_1 \tag{9}$$

What happened to the index: β was summed over (0,1,2,3), all four terms combined, and the result is $-2\not{p}_1$ which has **no** β **index**. The contracted pair has been consumed.

Substitute back into the trace:

$$= \frac{e^4}{4s^2}\,(-2)\,\mathrm{Tr}\Big[\gamma^\alpha\,(\not{p}_1+\not{p}_2)\,\not{p}_1\,(\not{p}_1+\not{p}_2)\,\gamma_\alpha\,\not{p}_4\Big] \tag{10}$$

Only one contracted pair remains: α.

9.2 Step 2: Simplify the Central Product

Before contracting α, simplify $(\not{p}_1 + \not{p}_2)\not{p}_1(\not{p}_1 + \not{p}_2)$ using $\not{p}_1^2 = 0$ and $\not{p}_2^2 = 0$.

$$\begin{aligned}(\not{p}_1 + \not{p}_2)\not{p}_1(\not{p}_1 + \not{p}_2) &= \underbrace{\not{p}_1\not{p}_1}_{=0}(\not{p}_1 + \not{p}_2) + \not{p}_2\not{p}_1(\not{p}_1 + \not{p}_2) \\ &= \not{p}_2\underbrace{\not{p}_1\not{p}_1}_{=0} + \not{p}_2\not{p}_1\not{p}_2 \\ &= \not{p}_2\not{p}_1\not{p}_2 \end{aligned} \tag{11}$$

Now use the anticommutation relation $\not{a}\not{b} = 2(a \cdot b) - \not{b}\not{a}$:

$$\begin{aligned}\not{p}_2\not{p}_1\not{p}_2 &= (2p_1 \cdot p_2 - \not{p}_1\not{p}_2)\not{p}_2 \\ &= 2(p_1 \cdot p_2)\not{p}_2 - \not{p}_1\underbrace{\not{p}_2\not{p}_2}_{=0} \\ &= 2(p_1 \cdot p_2)\not{p}_2 = s\not{p}_2 \end{aligned} \tag{12}$$

Index Note on the Anticommutation Step

When we write $\not{p}_2\not{p}_1 = 2(p_1 \cdot p_2) - \not{p}_1\not{p}_2$, each $\not{p}$ is $p_\mu\gamma^\mu$ with its *own* internally summed index. Expanding:

$$\not{p}_2\not{p}_1 = (p_{2\mu}\gamma^\mu)(p_{1\nu}\gamma^\nu) = p_{2\mu}p_{1\nu}\gamma^\mu\gamma^\nu$$

The anticommutation $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ then gives:

$$p_{2\mu}p_{1\nu}\gamma^\mu\gamma^\nu = p_{2\mu}p_{1\nu}(2g^{\mu\nu} - \gamma^\nu\gamma^\mu) = 2p_{2\mu}p_1^\mu - p_{1\nu}p_{2\mu}\gamma^\nu\gamma^\mu = 2(p_1 \cdot p_2) - \not{p}_1\not{p}_2$$

The dummy indices inside each $\not{p}$ are spectators — they get summed over independently and do not interfere with the external contracted indices α and β.

The trace is now:

$$= \frac{e^4}{4s^2}(-2)(s)\,\mathrm{Tr}\Big[\gamma^\alpha\,\not{p}_2\,\gamma_\alpha\,\not{p}_4\Big] \tag{13}$$

9.3 Step 3: Outer contraction (α)

Now the α pair has only $\not{p}_2$ between them:

$$\gamma^\alpha\ \underbrace{\not{p}_2}_{\text{1 slash}}\ \gamma_\alpha$$

Theorem 7 again: $\gamma^\alpha\not{p}_2\gamma_\alpha = -2\not{p}_2$.

$$= \frac{e^4}{4s^2}(-2)(s)(-2)\,\mathrm{Tr}\Big[\not{p}_2\,\not{p}_4\Big] = \frac{e^4}{s}\,\mathrm{Tr}(\not{p}_2\not{p}_4) \tag{14}$$

Both contracted pairs (α and β) have been consumed. No free Lorentz indices remain in the trace.

9.4 Step 4: Evaluate the Final Trace

$\mathrm{Tr}(\not{p}_2\not{p}_4)$ — expand:

$$\mathrm{Tr}(\not{p}_2\not{p}_4) = \mathrm{Tr}(p_{2\mu}\gamma^\mu\, p_{4\nu}\gamma^\nu) = p_{2\mu}p_{4\nu}\,\mathrm{Tr}(\gamma^\mu\gamma^\nu)$$

By Theorem 12: $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$. So:

$$= p_{2\mu}p_{4\nu}\cdot 4g^{\mu\nu} = 4\,p_{2\mu}p_4^\mu = 4\,p_2\cdot p_4 = 4\left(-\frac{t}{2}\right) = -2t$$

s-Channel Result

$$\boxed{\frac{1}{4}\sum|\mathcal{M}_s|^2 = \frac{e^4}{s}\times(-2t) = -\frac{2e^4\,t}{s}} \tag{15}$$

10 The t-Channel: By Photon Exchange $p_2 \to -p_3$

Replace $p_2 \to -p_3$ in the s-channel trace (8). Only the propagator and denominator change:

$$\frac{1}{4}\sum|\mathcal{M}_t|^2 = \frac{e^4}{4t^2}\,\mathrm{Tr}\Big[\gamma^\alpha\,(\not{p}_1-\not{p}_3)\,\gamma^\beta\,\not{p}_1\,\gamma_\beta\,(\not{p}_1-\not{p}_3)\,\gamma_\alpha\,\not{p}_4\Big] \tag{16}$$

The index structure is identical. The same contraction theorems apply:

1. $\gamma^\beta\not{p}_1\gamma_\beta = -2\not{p}_1$ (Theorem 7, same as before).
2. Central: $(\not{p}_1-\not{p}_3)\not{p}_1(\not{p}_1-\not{p}_3) = \not{p}_3\not{p}_1\not{p}_3 = 2(p_1\cdot p_3)\not{p}_3 = (-t)\not{p}_3$.
3. $\gamma^\alpha\not{p}_3\gamma_\alpha = -2\not{p}_3$ (Theorem 7).
4. $\mathrm{Tr}(\not{p}_3\not{p}_4) = 4\,p_3\cdot p_4 = 2s$.

Collecting: $\dfrac{e^4}{4t^2}\times(-2)\times(-t)\times(-2)\times 2s = -\dfrac{2e^4 s}{t}$.

t-Channel Result

$$\boxed{\frac{1}{4}\sum|\mathcal{M}_t|^2 = -\frac{2e^4\,s}{t}} \tag{17}$$

11 The Interference Trace: Index Tracking

$$\frac{1}{4}\sum\mathcal{M}_s\mathcal{M}_t^* = \frac{e^4}{4st}\,\mathrm{Tr}\Big[\gamma^\alpha\,(\not{p}_1+\not{p}_2)\,\gamma^\beta\,\not{p}_1\,\gamma_\alpha\,(\not{p}_1-\not{p}_3)\,\gamma_\beta\,\not{p}_4\Big] \tag{18}$$

Index Difference from $|\mathcal{M}_s|^2$

Compare (18) with (8). The **positions of α and β** have changed:

- In $|\mathcal{M}_s|^2$: β pair is inner (positions 3 and 5), α pair is outer (positions 1 and 7).
- In the interference: α pair is at positions 1 and 5, β pair is at positions 3 and 7.

The contracted pairs are **interleaved** ($\alpha\cdots\beta\cdots\alpha\cdots\beta$), not nested ($\alpha\cdots\beta\cdots\beta\cdots\alpha$). This

means we cannot do one contraction in isolation — we need Theorem 9 first.

11.1 Step 1: Contract α using Theorem 9

The α pair has three matrices between them: $(\not{p}_1 + \not{p}_2)$, γ^β, $\not{p}_1$.

Theorem 9: $\gamma^\alpha \not{a}\not{b}\not{c}\gamma_\alpha = -2\not{c}\not{b}\not{a}$.

$$\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \not{p}_1 \gamma_\alpha = -2\,\not{p}_1\,\gamma^\beta\,(\not{p}_1 + \not{p}_2) \tag{19}$$

Note: α is gone. β survives (it was not part of the contracted pair).

11.2 Step 2: Expand and contract β

The trace becomes:

$$\frac{e^4}{4st}(-2)\ \mathrm{Tr}\Big[\not{p}_1\,\gamma^\beta\ \underbrace{(\not{p}_1 + \not{p}_2)(\not{p}_1 - \not{p}_3)}_{\text{expand}}\ \gamma_\beta\ \not{p}_4\Big]$$

Expanding (using $\not{p}_1^2 = 0$):

$$(\not{p}_1 + \not{p}_2)(\not{p}_1 - \not{p}_3) = -\not{p}_1\not{p}_3 + \not{p}_2\not{p}_1 - \not{p}_2\not{p}_3$$

Apply Theorem 8′ ($\gamma^\beta \not{a}\not{b}\gamma_\beta = 4(a\cdot b)$) to each term:

$$\gamma^\beta(-\not{p}_1\not{p}_3)\gamma_\beta = -4(p_1\cdot p_3) = -4(-t/2) = +2t \tag{20}$$

$$\gamma^\beta(+\not{p}_2\not{p}_1)\gamma_\beta = +4(p_2\cdot p_1) = +4(s/2) = +2s \tag{21}$$

$$\gamma^\beta(-\not{p}_2\not{p}_3)\gamma_\beta = -4(p_2\cdot p_3) = -4(-u/2) = +2u \tag{22}$$

Sum: $2t + 2s + 2u = 2(s+t+u) = 0$.

Interference = 0

$$\boxed{\frac{1}{4}\sum \mathcal{M}_s\mathcal{M}_t^* = 0} \tag{23}$$

The interleaved index structure forces us through Theorems 9 and 8′, which distill the 8-γ trace down to the scalar $2(s+t+u)$, and kinematics kills it.

12 Final Answer

Compton $\overline{|\mathcal{M}|^2}$

$$\boxed{\overline{|\mathcal{M}|^2} = -\frac{2e^4 t}{s} - \frac{2e^4 s}{t} = -2e^4\left(\frac{t}{s} + \frac{s}{t}\right)} \tag{24}$$

Appendix: Index Cheat Sheet

Quick Reference

Free index: Appears once. Labels a family of equations. NOT summed.
Contracted (dummy) index: Appears twice (once upper, once lower). IS summed.
Rule: Every Lorentz index must be either free or contracted. Never three of the same index.

Where indices come from in this problem:

- α, ρ: from photon 3 polarization ε_3 and the γ-matrix at its vertex.
- β, σ: from photon 2 polarization ε_2 and the γ-matrix at its vertex.
- Internal dummies inside each $\not{p} = p_\mu \gamma^\mu$: summed on creation, invisible.

What the contraction theorems do: They take a contracted pair $\gamma^\mu \cdots \gamma_\mu$ and **consume** it, reducing the number of γ-matrices in the trace by 2 (or converting them to dot products/numbers).